

Sex-Linked inheritance:

Sex linked genes are those located on either the X or Y chromosomes. Because few genes are known to be located on the human Y chromosome, our attention will be focused on X-linked diseases.

X—linked recessive inheritance:

-Color blindness(color vision deficiency):

-In human, there are three genes involved in distinguishing color and there are three different types of cones.

-The gene coding for the pigment responsive to blue light is on an autosome(chromosome 7).

-the genes coding for the pigments responsive to red and green are both on the X chromosome.

-In males if the gene for green is defective, green cannot be distinguished from red and conversely, a defect in the gene for red result in red appearing as green.

-In heterozygous females, the defective alleles are recessive to the normal alleles on the other X chromosome and so vision is usually normal.

$X^C X^C$ (Normal color vision female)

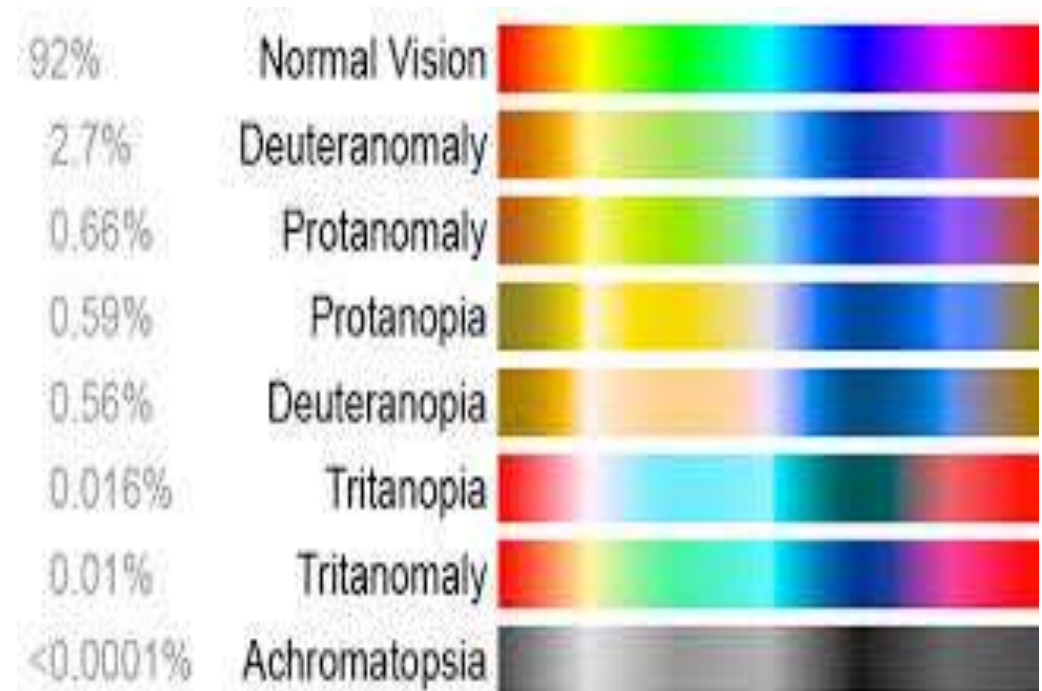
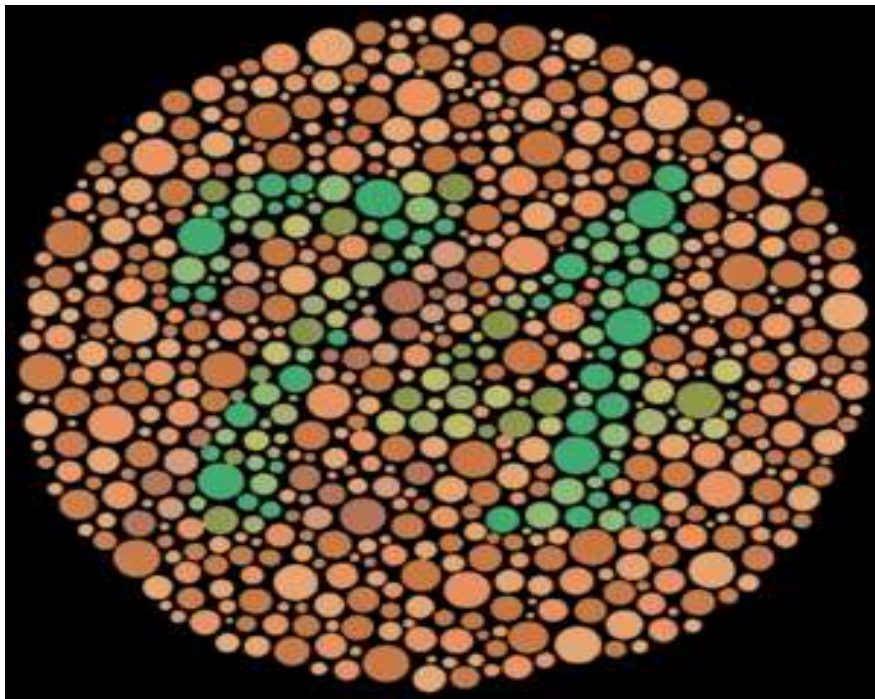
$X^C X^c$ (female carrier for the color vision deficiency)

$X^c X^c$ (Color blind female)

$X^C Y$ (Normal color vision male)

$X^c Y$ (color blind male)

- Complete red-green color blindness in females occurs only if both X chromosomes carry the same defective alleles .
- Any male receiving the recessive alleles from his mother will express the trait since, lacking a second X chromosome.



Ishihara color test plate : No. 74 or 21 ?

Charts show how different colorblind people see compared to a person with normal color vision.

Color blind persons are not really color blind since they can still perceive a large different array of colors but they are unable to see in a normal way(can not distinguishing between certain colors).

Types of color blindness

-Monochromatic: Either no cones available or just one type of them (rod monochromacy or achromatopsia)is a total color blind, everything appeared black and white or as shading of gray .

-Dichromatic: is caused by missing one of the three retinal cones

1-protanopia: complete absence of red cone but with normal green and blue cones(difficulties in distinguishing between red and green colors.

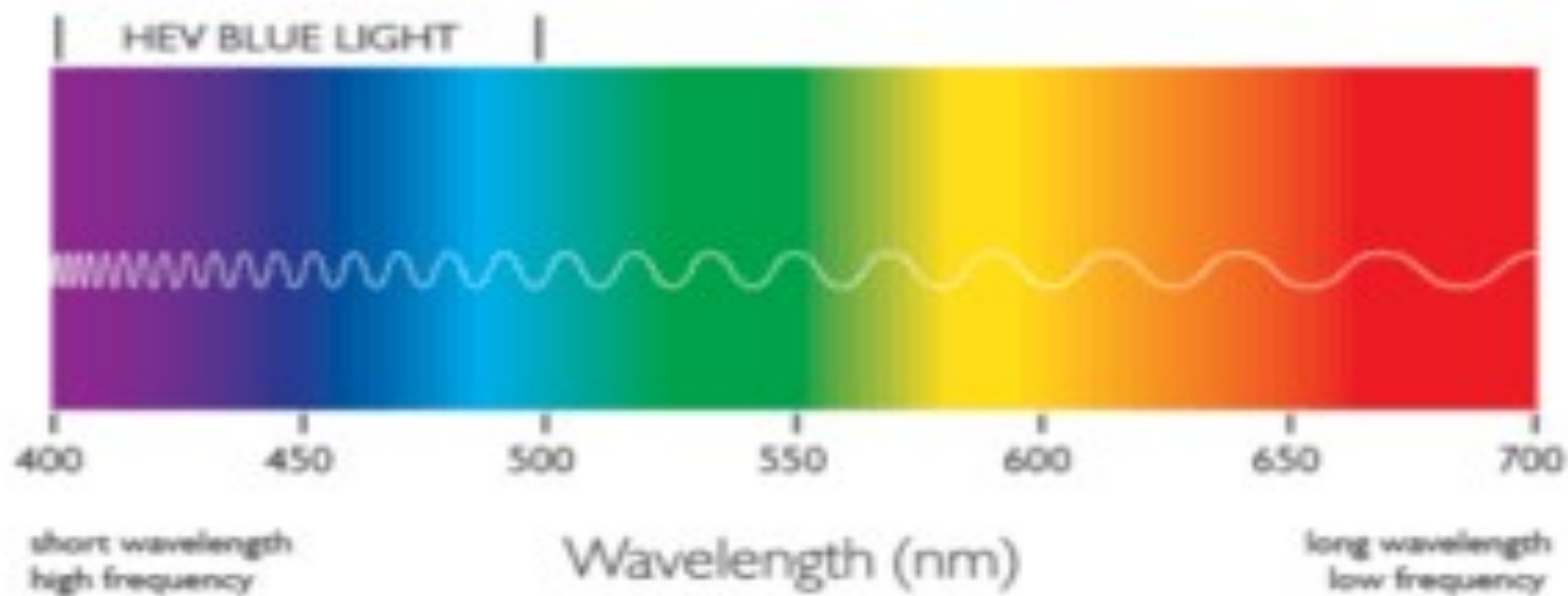
2-deutanopia: complete absence of green cone but with normal red and blue cones(green is the main problem colors , some gray, purple, greenish blue cannot be distinguished very well.

3- Titanopia: complete absence of blue cone but with normal red and green cones(blues appear greenish, yellows and oranges appear pinkish).

Anomalous trichromacy is a mild or weak color vision deficiency, occurring when one of the three cone pigments is not absent but altered or shifted in its spectral sensitivity(they have reduced sensitivity to distinguishing different shading of red ,green and blue colors).

VISIBLE LIGHT SPECTRUM CHART

NATURAL VISIBLE LIGHT SPECTRUM IN NANOMETERS



Hemophilia (a sex-linked recessive disease)

- Hemophilia A, or classical hemophilia, affects 1\10,000 males. It is the most common of the severe bleeding disorders.
- A high frequency of hemophilia has afflicted some royal families of Europe (Queen Victoria of England).
- Hemophiliacs bleed excessively when injured, because they have missing a certain plasma protein, known as **factor VIII**.

- Fibrin is failed to produce, resulting in prolonged and severe bleeding from wounds, even minor injuries carry the risk of the patients bleeding to death.
- Persons with hemophilia A can be treated with clotting factor VIII extracted from normal human blood, but the cost is very high.
- Currently a genetically engineered factor- **VIII** can be synthesized in bacteria.

Different types of hereditary bleeding disorders:

- Hemophilia A** caused by the deficiency of **factor VIII**
- Hemophilia B** occurs due to the missing or deficiency of **factor IX**
- Hemophilia C** caused by the deficiency of **factor XI**
- Parahemophilia** occurs due to the deficiency of **factor V**
- Von willebrand disease** caused by missing or defective factor (VWF)

Categories of hemophilia A:

1-**Severe** :In which the patients will produce less than 1% of Factor VIII in their blood. These individual experience frequent bleeding episodes spontaneously, often several per month.

2- **Moderate**: They produce about 5 % of Factor VIII and experience frequent bleeding episodes, mostly several per year after mild trauma.

3-**Mild**: They produce about 25-45 % of Factor VIII and experience bleeding episode only after surgery or severe trauma.

-Muscular dystrophy:

-Duchene muscular dystrophy affects about one in every 5,000 male births. The signs and symptoms typically appear in early childhood usually before the age 5. As the name implies, is characterized by a progressive wasting of the muscles and loss of coordination.

-All skeletal muscle degenerates eventually. The heart and respiratory muscle become impaired and death usually results from respiratory and cardiac failure.

-Recently the gene for DMD was isolated and it was discovered that the absence of a protein now called **dystrophin** is the cause of the disorder.

-The X-linked recessive allele remains in the population by passage from carrier mother to carrier daughter because death usually occurs by age twenty therefore affected males are rarely fathers.

Ichthyosis (fish scales disease)

By a year of age the affected boy would presented with rough, brown, scaly skin. An enzyme deficiency (steroid sulfatase) blocks removal of cholesterol from the skin cells.

The upper skin layer cannot peel off as it normally does.



X-Linked Dominant Inheritance:

- X-linked dominant diseases are fewer in number and less clinically significant than are X-linked recessive diseases.
- An example of X-Ldd is hypophosphatemic rickets, a disease in which the kidneys are impaired in their ability to reabsorb phosphate. This results in abnormal ossification with bending and distortion of the bones and fractures.

-The **fragile X syndrome(Martin-Bell syndrome)**

is an X-linked dominant condition seen in approximately 1\4000 males and 1\8000 females.

-It is characterized by a distinctive facial appearance, with large ears and long face , protruding chin, hyper mobile joints and macroorchidism in post pubertal males.

-The degree of mental retardation tends to be milder and more variable in females than in males



Congenital generalized hypertrichosis (hair overgrowth):

Affected boy produces many extra hair follicles, and hence denser and more abundant upper body hair.

Hair growth is milder and patchier in females because of hormonal differences and the presence of a second X chromosome



Y-Linked Inheritance (holandric inheritance):

- The Y chromosome contains relatively few genes.
- These include the gene that initiates differentiation of the embryo into a male, several genes encoding testis-specific spermatogenesis factors and minor histocompatibility antigens termed H_y, are located on the Y chromosome.
- Transmission of Y-linked traits is strictly father-son(e.g. Hairy pinna).

Sex-Limited and Sex-Influenced Traits.

- A sex-limited trait occurs in only one of the sexes, due to anatomical differences, for example inherited uterine defects in female and inherited testicular defects in male.
- A good example of a sex-influenced trait is male-pattern baldness which occurs in both males and females, but much more commonly in males.

-It is thought to be inherited as an autosomal dominant trait in males, where as in females it is inherited as an autosomal recessive trait.

-Females display the trait only when they inherit two copies of the alleles (homozygous), but heterozygous female with adrenal tumors develop pattern baldness due to an abnormality large amount of testosterone